

Mapping the 3D Parton Structures of Hadrons with Lattice QCD

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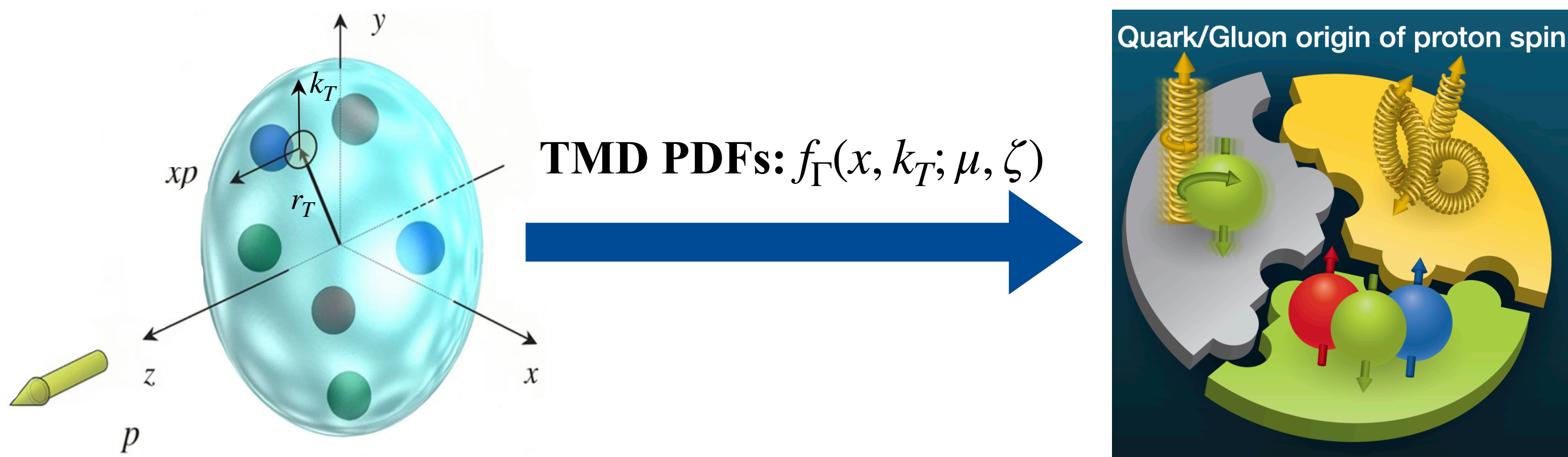
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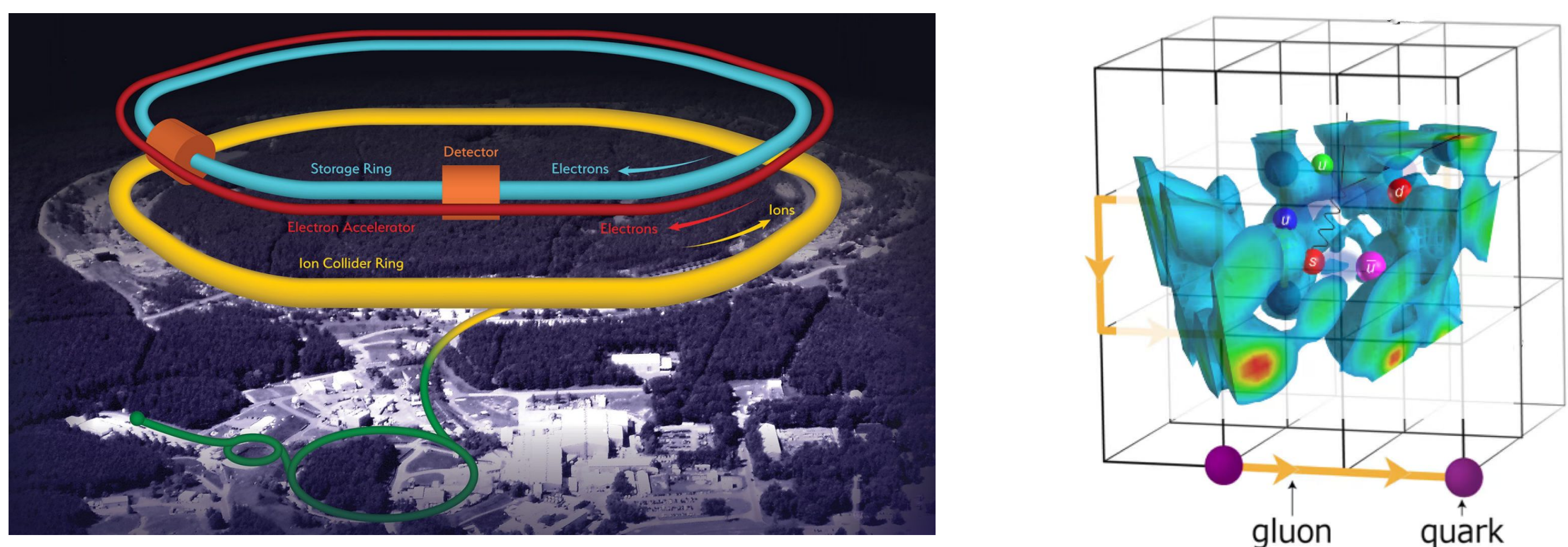
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3D Parton Structures in EIC Era

Parton distribution functions (PDF) and transverse momentum distributions (TMDs) encode the multidimensional partonic structure of hadrons, including the momentum, spin, and polarization correlations of quarks and gluons.

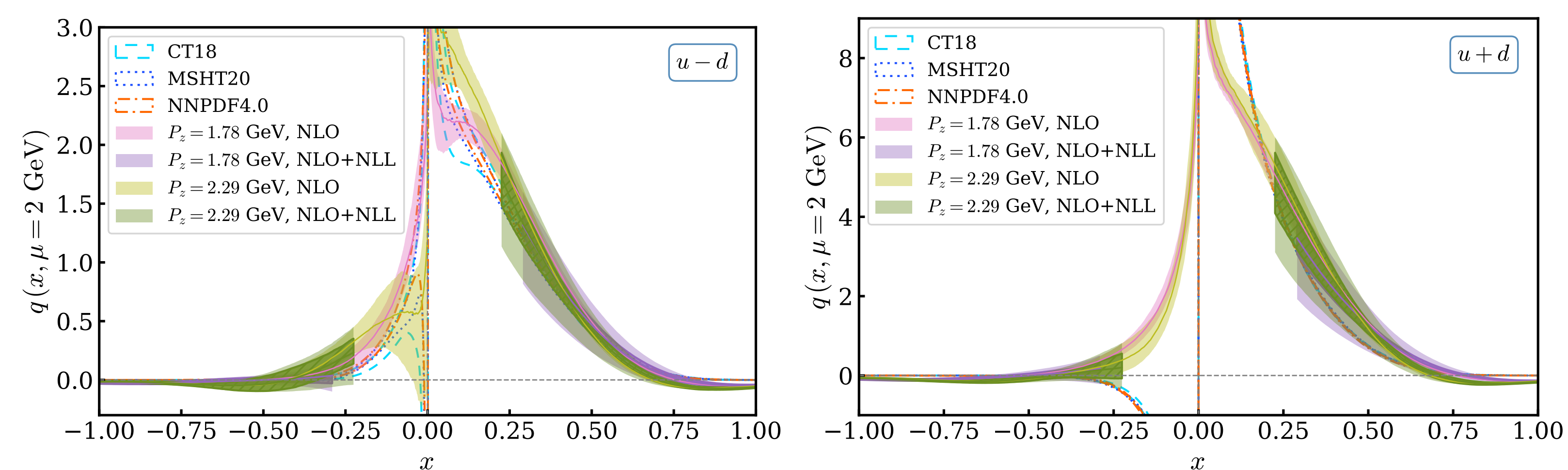


Lattice QCD provides **first-principles** predictions of hadron structure that complement and guide EIC measurements to reveal the **multidimensional quark and gluon dynamics** inside hadrons.



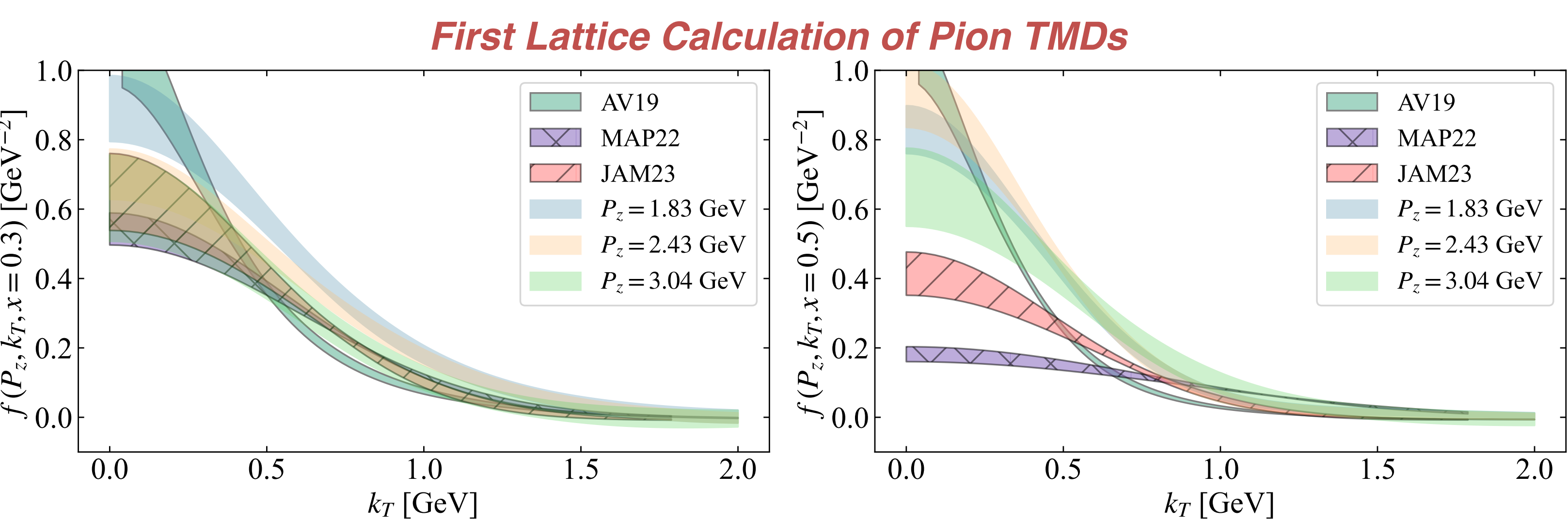
Proton Unpolarized PDF at the Physical Point

- **Physical-point** lattice QCD calculations up to $P_z = 2.29$ GeV yield **convergent** proton PDFs that agree with global analyses in the **moderate- x region**.



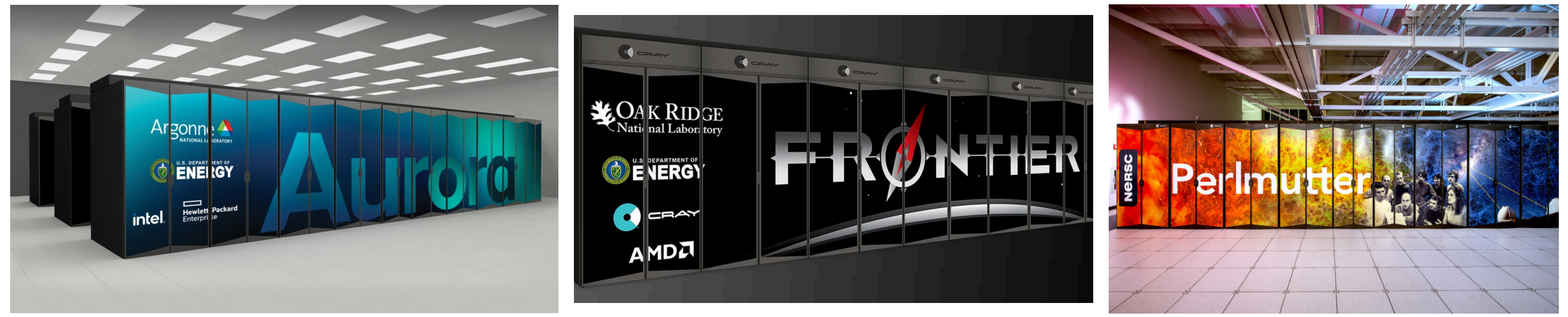
Pion Unpolarized TMDs on a Fine Lattice

- **Lattice provides predictions at relatively large x , where phenomenological results have large discrepancies with each other.**

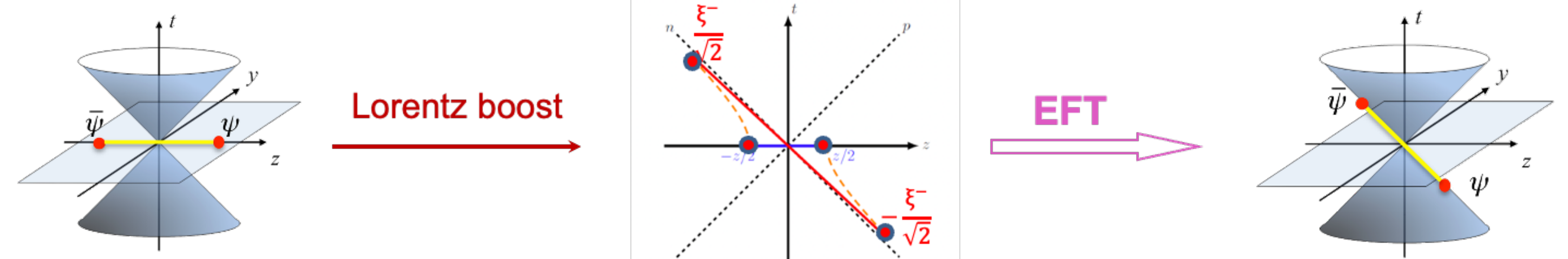


First-principles calculation for EIC physics

Our project has utilized the state-of-the-art exascale HPC facilities across the country, including Aurora, Frontier, and Perlmutter.



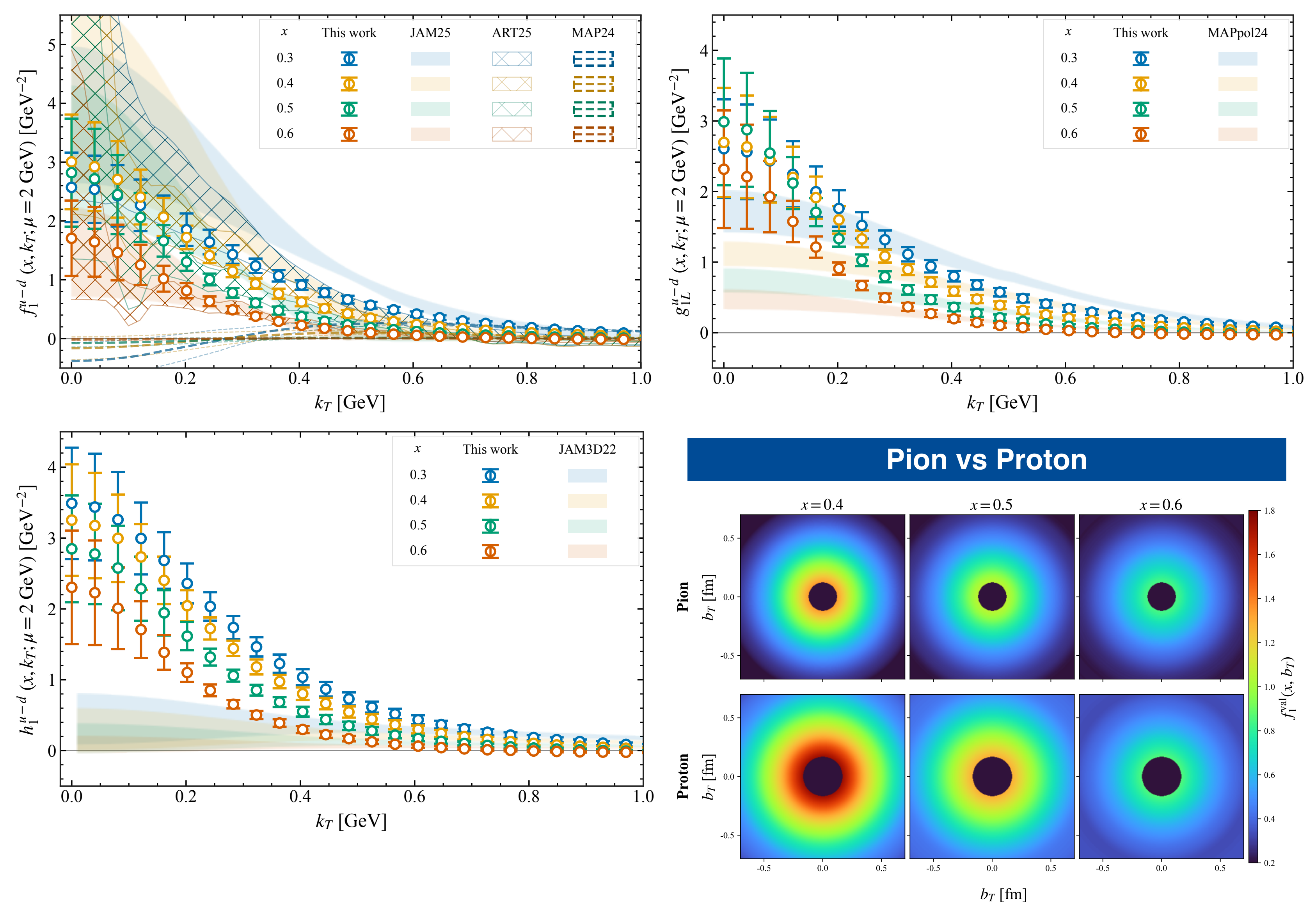
Large Momentum Effective Theory (LaMET) enables precision-controlled determinations of x -dependent parton distributions from lattice QCD.



Proton TMDs at the Physical Point

- We simultaneously obtain the unpolarized, helicity and transversity TMDs.
- Tensions among global analyses highlight the importance of **first-principles** lattice calculations as an **independent** determination of TMDPDFs.

First Simultaneous Lattice Calculation of Proton TMDs in Three Polarizations



Other Research Highlights

- **Quark Collins-Soper kernel;**
- **Pion generalized parton distributions (GPDs);**
- **Pion & Kaon PDFs and light-cone distribution amplitudes (LCDAs);**
- **Pion & Kaon electric-magnetic form factors at large momentum transfer;**
- **Hadrons and light nuclei masses and trace anomaly;**

References

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